

The game reads world generation data from data packs or world_gen_settings.dat inconsistently, preventing updating large biome sources

MC-307144 [Save Data](#)

Status	Resolved / Fixed
Affected	26.1, 26.1.1, 26.1.2, 26.2 Snapshot 2, 26.2 Snapshot 3, 26.2 Snapshot 5
Fixed in	26.2 Snapshot 7
Reported	2026-03-30

STEPS TO REPRODUCE

1. Launch Minecraft 1.21.11.
2. Install the 1.21.11 version of the Terralith data pack.
3. Create a new world with Terralith installed.
4. Save and exit the world.
5. Quit the game and upgrade the instance to Minecraft 26.1.
6. In the test world's files, delete the 1.21.11 version of Terralith and replace it with the 26.1 version.
7. Launch the 26.1 instance, open Singleplayer, select the test world, and click "Upgrade and Play".
8. Attempt to join the upgraded world.
9. Observe that the game prevents joining the world because of errors in the data pack or level data.

OBSERVED BEHAVIOUR

Actual behavior

In Minecraft 26.1, world-generation settings are handled inconsistently depending on how the world was created or migrated:

- For a newly created 26.1 world, the contents of `world\_gen\_settings.dat` do not necessarily reflect the dimensions and biome sources supplied by installed data packs. For example, with Terralith and New in Town: Dimensions installed, the file retains the vanilla Overworld preset and does not list New in Town's custom dimensions, even though both data packs function correctly in game.
- `world\_gen\_settings.dat` is nevertheless read when it contains manually edited settings. For example, replacing its Overworld biome source with a list containing only `minecraft:plains` causes the world to generate exclusively Plains biomes.
- When a 1.21.11 world is upgraded to 26.1, the complete `WorldGenSettings` data from `level.dat` is migrated into `world\_gen\_settings.dat`. This includes large custom biome-source lists that are not written to the file when an equivalent world is created directly in 26.1.
- If the migrated biome-source data exceeds the NBT reader's limit, Minecraft fails to load the world with an `NbtAccounterException`:

```
```text
```

```
Tried to read NBT tag that was too big; tried to allocate: 2097146 + 8 bytes where max allowed: 2097152
```

```
```
```

The failure prevents the player from joining the world and results in an error indicating that the data pack or level data is invalid. Deleting `world\_gen\_settings.dat` does not resolve the issue: the large biome-source data is written back into the file, rather than being replaced with a usable set of 26.1 defaults. This leaves upgraded worlds with sufficiently large custom biome sources unable to load, even when the corresponding data pack works correctly in a newly created 26.1 world.

EXPECTED BEHAVIOUR

When upgrading the 1.21.11 world to 26.1 with the updated Terralith datapack, the world should load and remain playable without errors. Its custom biome source and other datapack-defined world generation settings should be

preserved and applied consistently, regardless of whether the world is newly created or migrated. Large biome data should not prevent loading, and the existing world seed and generation should remain unchanged.

ENVIRONMENT

Minecraft Java Edition Versions: 26.1, 26.1.1, 26.1.2, 26.2 Snapshot 2, 26.2 Snapshot 3, 26.2 Snapshot 5 (includes snapshots)

ORIGINAL DESCRIPTION

In 26.1, level.dat was split up, with WorldGenSettings being moved to its own world_gen_settings.dat file. However, the behaviour of this data has changed, which causes issues with updating worlds that have large custom biome sources.

1. Datapacks no longer affect worldgen settings

For 1.21.11, WorldGenSettings in the level.dat would store data regarding dimensions that were edited by a datapack. See Image 1 to view how this looks when Terralith and New in Town: Dimensions are installed. Terralith replaces the overworld preset with the full biomes list supplied by its dimension/overworld.json file, and all of New in Town's dimensions are registered.

In 26.1, this is no longer the case - see Image 2 to view what world_gen_settings.dat looks like with the exact same datapacks installed. The overworld's preset is still set and not replaced with Terralith's biome source; and none of New in Town's dimensions are registered here either. Both Terralith and New in Town: Dimensions work in game though, so I assume the game reads directly from the datapack instead of copying the data to world_gen_settings.dat.

It's worth noting that although world_gen_settings.dat does not seem to match exactly what is happening in game, it is not completely ignored. When I created a fully Vanilla world, and then edited the world_gen_settings.dat by hand to replace the preset with a custom biomes list that consisted of a single minecraft:plains entry, re-entering the world did correctly spawn only Plains biomes, as expected.

2. Migrating from 1.21.11 to 26.1 preserves worldgen settings

Normally this seems like intended and expected behaviour, but because of the previous section, this causes inconsistencies. In Image 3, it shows what the world_gen_setting.dat file looks like when migrating a 1.21.11 world (the exact same one used for Image 1) to 26.1. Notice how the biomes list is still present. Although the data is preserved correctly through migration, it is different than Image 2, in which the world has the exact same setup, but instead was created in 26.1 instead of being migrated from 1.21.11.

There is also an odd inconsistency when the world_gen_settings.dat is deleted.

MANUAL RESULT

| | |
|------------------|--------------------|
| Version used | |
| Reproduced? | YES / NO / PARTIAL |
| Crash file | |
| Notes | |
| | |
| Tested by / date | |

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-307144>